MIL-S-10203A (Ord)
3 August 1955

SUPERSEDING
MIL-S-10203 (Ord)
22 March 1950

MILITARY SPECIFICATION
SYSTEM, SIGHTING, T34E1

1. SCOPE

1.1 This specification covers one type of sighting system known as System, Sighting T34E1. This sighting system is part of Fire Control System T38, used with 75mm Gun T83E1 and 75mm AA Gun Mount T69.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings and those specifications and drawings referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military	
MIL-T-11792	Telescope, T155 and T155E1
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps	
F7630292	System, Sighting, T34E1

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements of Specification MIL-F-13926 shall be mandatory.

3.2 Material.- Material shall be in accordance with the drawings and specifications forming a part of this specification.

3.3 Telescopes.- Telescopes shall meet the requirements of Specification MIL-T-11792.

3.4 Alinement.- The line of sight of the telescopes shall be adjusted parallel and then adjusted perpendicular to the gun mounting surface of the main support bracket. This shall be done with the sighting system mounted on a suitable test fixture in the position as shown on the assembled view drawing. The line of sight of each telescope shall not deviate from the initial alinement by more than 0.5 mil, when the two telescope mount brackets are removed or folded out of position and repositioned.

3.5 Environmental

3.5.1 Temperature.- The sighting system shall not suffer damage or fail to meet the requirements of 3.4 when tested in accordance with 4.2.2.

3.5.2 Vibration.- The sighting system shall not suffer damage or fail to meet the requirements of 3.4 when tested in accordance with 4.2.3.

3.5.3 Shock.- The sighting system shall not suffer damage or fail to meet the requirements of 3.4 when tested in accordance with 4.2.4

3.6 Workmanship.- Workmanship shall be in accordance with Specification MIL-F-13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified, each sighting system shall conform with the quality assurance provisions of this specification and Specification MIL-F-13926.

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4.2 Environmental tests

4.2.1 General.- One of the first five sighting systems and one out of each additional fifty or fraction thereof shall be tested. A system that has satisfactorily complied with the requirements of 3.4 shall be subjected to all of the environmental tests.

4.2.2 Temperature.- A sighting system shall be placed in the chamber and exposed to an air temperature of minus 62°C (- 80°F) for a period of 3 days. The air temperature then shall be raised to minus 40°C (- 40°F) and after the temperature of the sighting system has reached equilibrium, it shall be tested for conformance with 3.4. Raise the air temperature to normal room temperature and test for conformance with 3.4. Raise the air temperature to plus 71°C (+ 160°F) and after the temperature of the sighting system has reached equilibrium at that temperature allow it to remain at that temperature for a period of 4 hours. Reduce the air temperature to plus 52°C (+ 125°F) and after the temperature of the sighting system has reached equilibrium at that temperature expose the top surface to 360 BTU/Ft. Sq/Hr., (50 to 60 percent of the total energy shall be in wave lengths above 8,000 angstrom units and 4.5 to 6.0 percent in wave lengths below 4,000 angstrom units), for a period of 4 hours. The sighting system then shall be tested for conformance with 3.4. Allow the unit to return to normal ambient temperature and test for conformance with 3.4.

4.2.3 Vibration.- The sighting system shall be mounted on the vibration table in a position that will simulate stowing. Each unit tested then shall be vibrated successively along each of the three mutually perpendicular directions with accelerations or excursions as shown below. The applied frequency shall be changed at a uniform rate between the limits indicated, and the rate of change shall not exceed one cycle per second. Test for conformance with 3.4.

Frequency limits	Excursions	Acceleration	Time (each direction)
0 - 10 cps	0.060 inch	- - - - -	1 hour
11 - 20 cps	0.030 inch	- - - - -	2 hours
21 - 400 cps	- - - - -	4.5 g's	2 hours

4.2.4 Shock.- The sighting system shall be mounted rigidly, using its own mounting provisions, to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of

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the three mutually perpendicular directions, making a total of 18 shocks. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured shock impulse. At the completion of this test the sighting system shall be tested for conformance with 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with instructions supplied by the mission arsenal through the contracting officer.

6. NOTES

6.1 Intended use.- Sighting System T34E1 is designed for use with the Skysweeper as an auxiliary, direct-fire system.

6.2 Ordering data.- Contracts and invitation for bids should specify the title, number, date of this specification and type of packaging and packing.

Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army - Ordnance Corps

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